

1. Sample questions based on **LECTURES** content

(Multiple-Choice) One of the problems of Natural Languages is that different sentences can have the same meaning. This is known as the problem of:

- a. Overfitting
- b. Ambiguity
- c. Variability
- d. Generalization

Answer: **c**

(Open, development question) Explain how the self-attention mechanism works inside a Transformer.

Answer: *The self-attention mechanism in a Transformer works by calculating attention scores between every pair of words in the input sequence. To compute these attention scores, three types of embeddings are generated for each word: query, key, and value. The attention scores are calculated by taking the dot product between the query and key embeddings, followed by applying a softmax function to obtain a probability distribution. These attention scores determine how much each word should contribute to the final representation of the current word. Finally, the attention-weighted sum of the value embeddings, weighted by the attention scores, is computed. This weighted sum serves as the contextual representation of the current word, capturing the relevant information from the entire sequence.*

2. Sample questions based on **LABS** content

(Multiple-Choice) What is the purpose of Lemmatization in Text Mining?

- a. Removing stop words from the text
- b. Reducing words to their root form
- c. Grouping words with similar meanings together
- d. Removing prefixes and suffixes from words

Answer: **b**

(Open, development question) II.5 Considering that you had to create and train a Natural Language Processing (NLP) pipeline to perform classification over a new corpus, define and explain what you think are the three (3) main steps of that NLP pipeline considering the employment of Transformers for embedding generation.

Answer:

1. **Preprocessing:** *This step involves cleaning and formatting the corpus data, which may include tasks such as removing special characters, lowercasing the text, and handling any specific requirements of the dataset.*
2. **Transformers (Embedding Generation):** *The tokenized input is passed through a pre-trained Transformer model, such as BERT, to generate contextualized sentence/word embeddings.*
3. **Classification:** *The aggregated representation of the input sequence is fed into a classification model, such as a fully connected neural network or a logistic regression classifier, which predicts the class or label for the given text.*